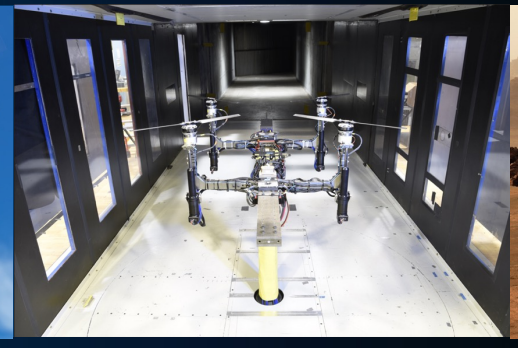
Status and Preliminary Results from NASA/Army Hover Test

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Hover Test Overview

- **Summary**
  - After a number of delays/setbacks, NASA and the U.S. Army were able to begin research testing of the HVAB rotor in the NFAC in the fall of 2022
- **Research Accomplishments**
  - Data acquired at 3 tip Mach numbers for multiple configurations
    - Standard blades – natural transition
    - Standard blades – forced transition (3.5 mil trip dots)
    - Pressure blade – natural transition (limited data)
  - Includes measurements of performance, blade deflection (photogrammetry), blade transition (IR thermography), rotor wake position (shadowgraphy), and blade airloads (blade pressures)
- **Status**
  - Data review and analysis is underway
    - Some additional performance validation and/or wake characterization (PIV) testing is under consideration
  - Test results to be presented at VFS Forum in May
  - Goal is to have validated data ready to distribute in May/June time frame

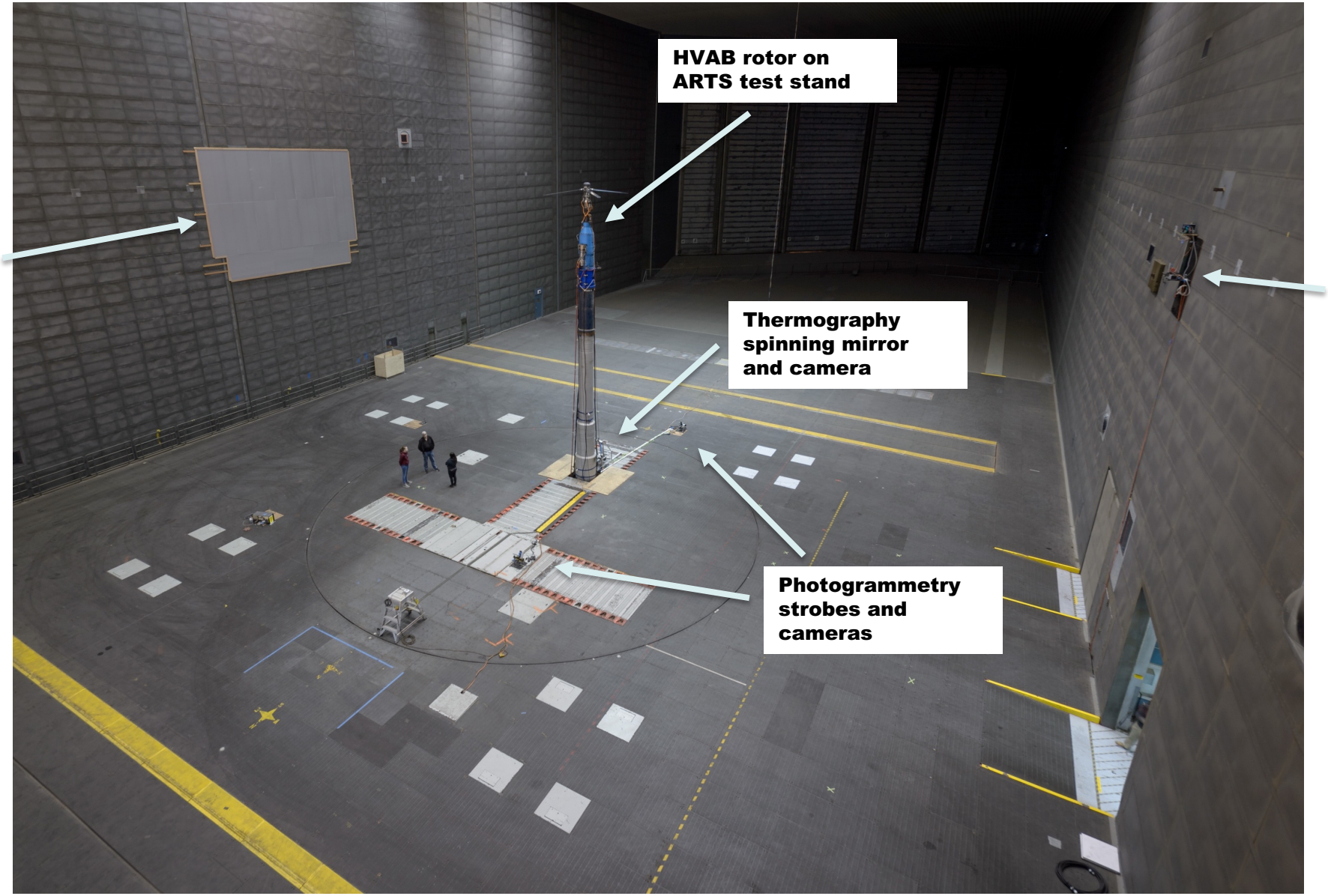
Planned Test Objective and Approach

- **Objective**
  - Acquire key experimental data for a rotor in hover to advance the SOA of simulations of hover performance in free air
- **Requirements**
  - Test measurements should be sufficiently accurate for CFD validation (e.g. FM = +- .005) and comprehensive enough to ensure correct physics are represented
  - Experimental uncertainties due to effects that aren't easily measurable (facility walls and Reynolds number) should be minimized
- **Approach**
  - Develop 4-bladed blade set based on Army's PSP rotor design with pressure and strain gage instrumentation
  - Test in NFAC 80- by 120-Foot test section to minimize facility effects on performance
  - Acquire key measurements for CFD validation
    - Performance, airloads, transition location, wake info, blade deflections, etc



HVAB rotor installed in 80x120

Wide-angle View of Test Installation



Shadowgraph retro-reflective screen

Shadowgraph laser and camera

Test Conditions and Data Acquired

Configuration	Mtip (or RPM)	Collective (deg)	Measurements	Primary Objective
Standard blades – natural transition	1160, 1250, 1310 RPM	4, 6, 8, 10, 11, 12, 13, 14	P, PG	Blade deflection
	0.60, 0.65, 0.675	4 to 15, 1 deg incr	P, IRT	Performance and transition
	0.60, 0.65, 0.675	8, 10, 12, 14	P, IRT, SG	Wake geometry
Standard blades – forced transition	0.60, 0.65, 0.675	4 to 15, 1 deg incr	P, IRT	Fully tripped 3.5 mil dots
	0.60, 0.65, 0.675	4, 6, 8 (0.65 only) 10, 12, 14 (all)	P, IRT	Tripped lower surface only
Pressure blade	0.60	4 to 12, 1 deg incr	P, IRT, limited SG	Blade Airloads
	0.65	4, 6, 8, 10, 11, 12, 13	P, IRT, limited SG	Blade Airloads

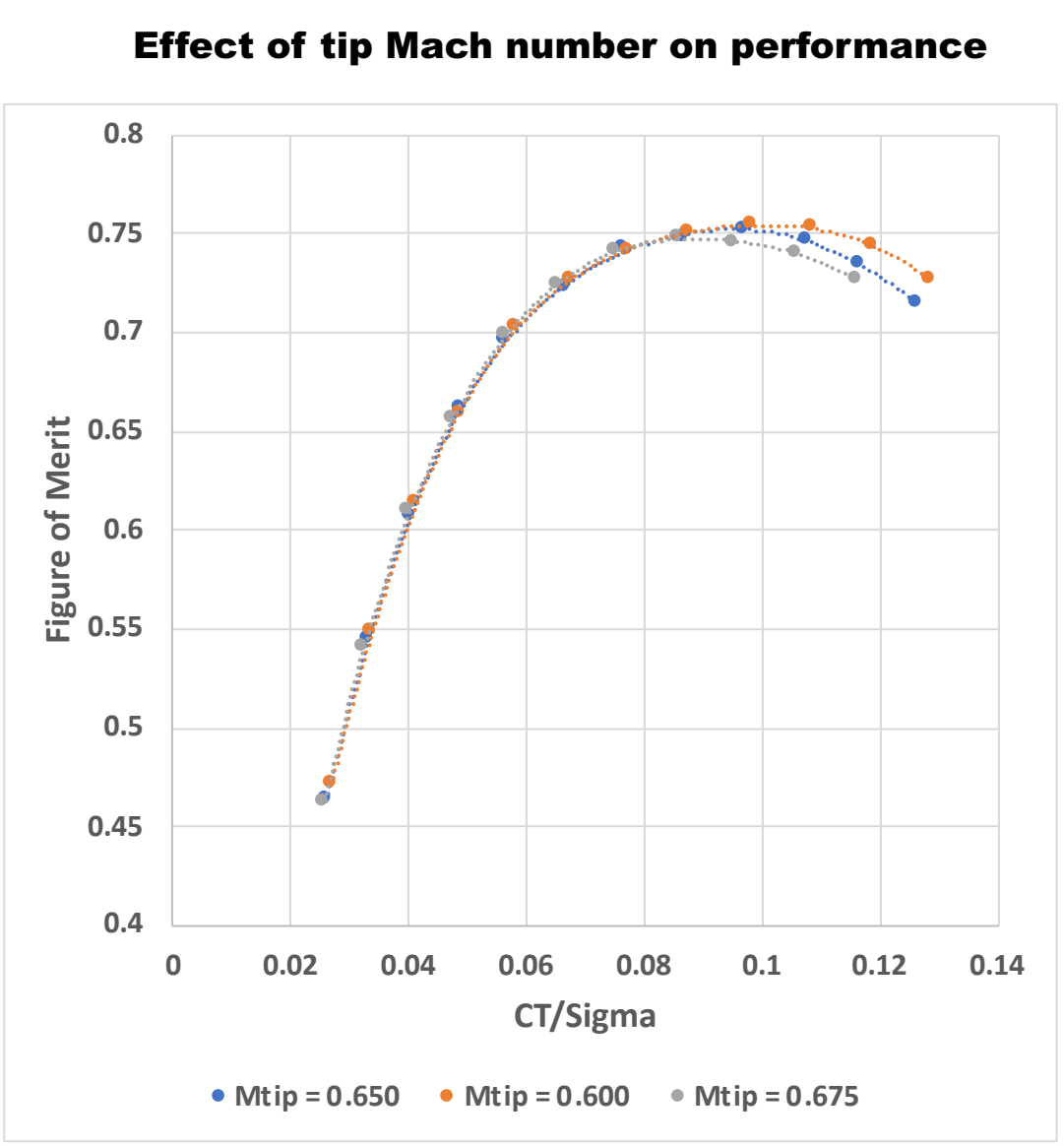
Performance (P), IR Thermography (IRT), Shadowgraphy (SG)

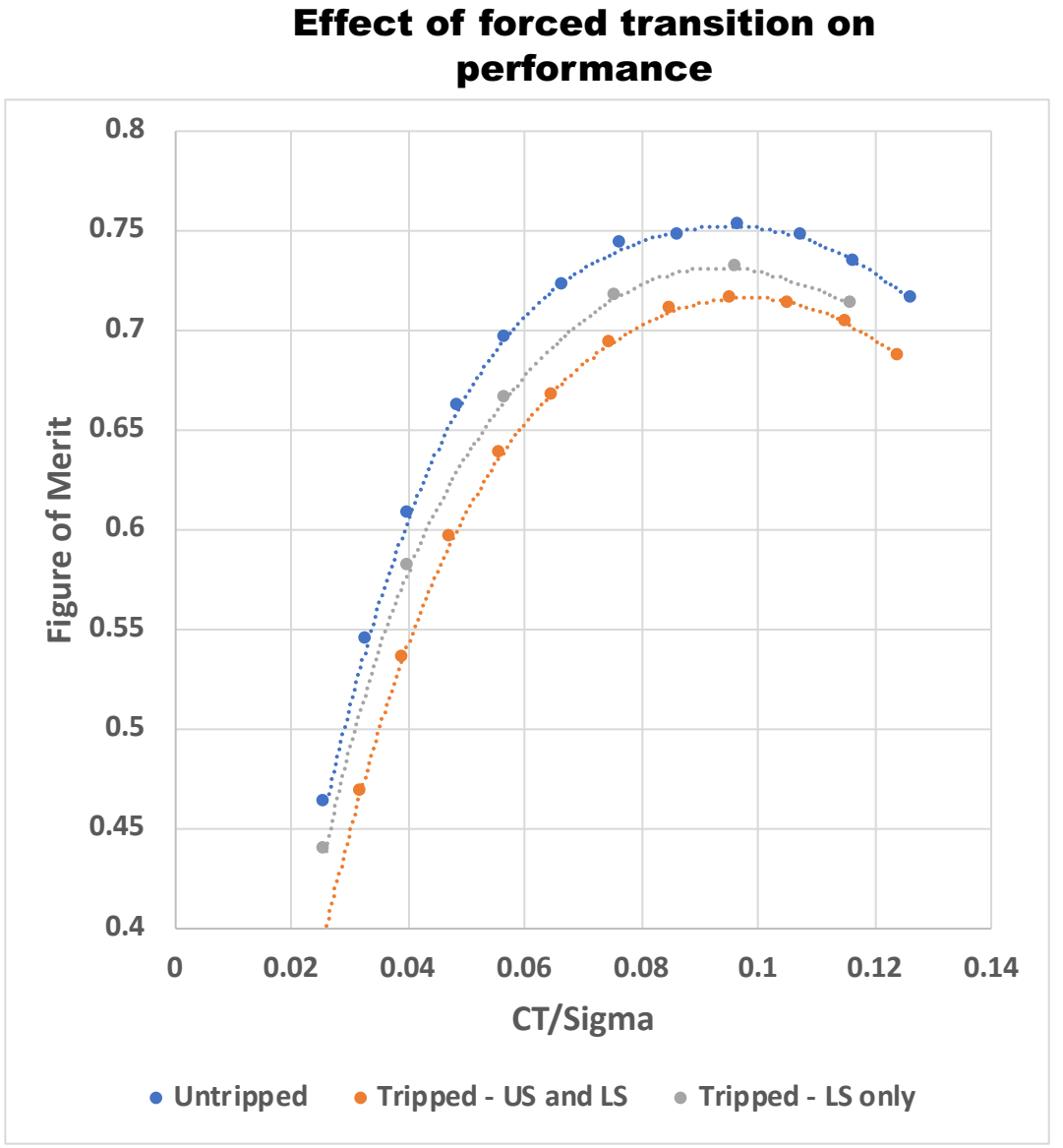
Testing Procedures and Observations

- **Accurate performance data required specific testing procedures**
  - Rotor balance (thrust) required 45-minute pre-heat run to minimize drift
  - Performance measurements showed low-frequency variability due to recirculation (approx. 60-sec period). Figure of Merit accuracies of 0.005 required averaging data over multiple data points (typically 10 points of 128 rev duration acquired every 30 sec).
    - Outside winds did not affect tunnel internal flow or recirculation
    - Photogrammetry and Shadowgraphy data were also averaged over longer periods to minimize variability

- **Pressure blade failure during testing limited airload data acquisition**
  - Failure was a known possibility based on earlier LaRC results
  - Prior to failure, research data were acquired at 2 Mtip values

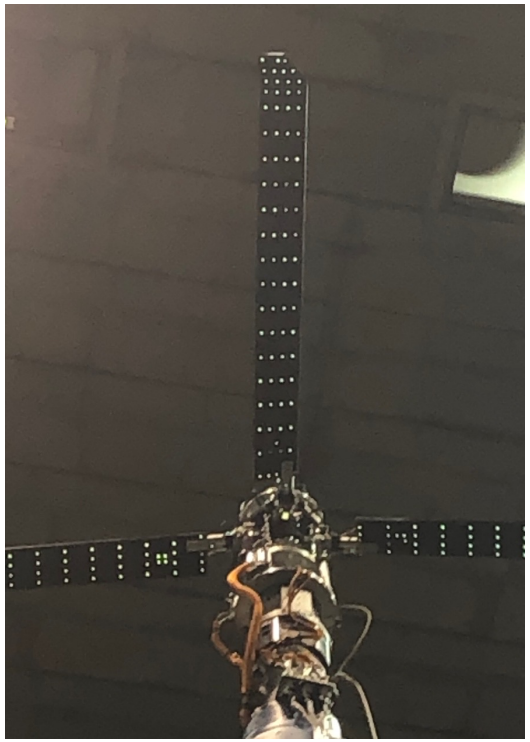
Sample Data – Performance of Standard Blades



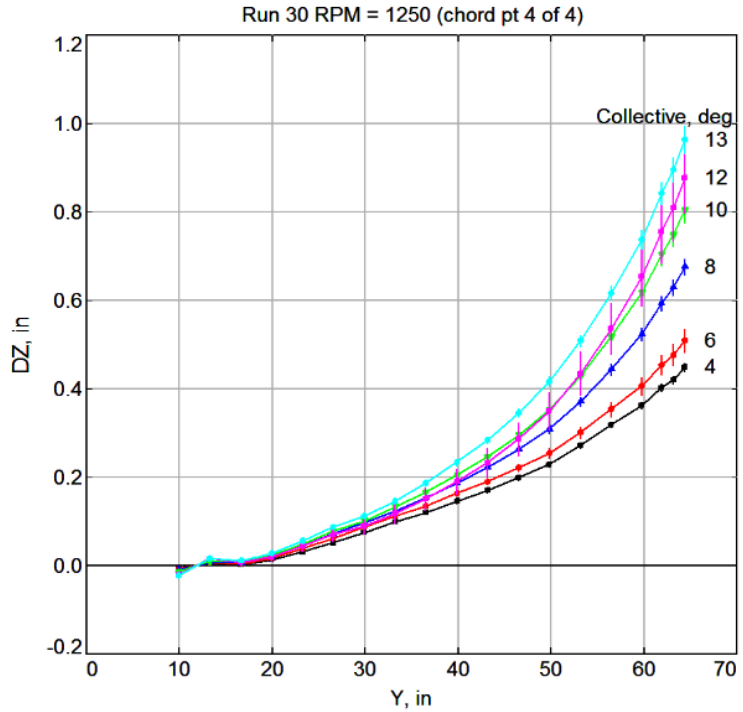


Sample Data – Blade Deflections (Photogrammetry)

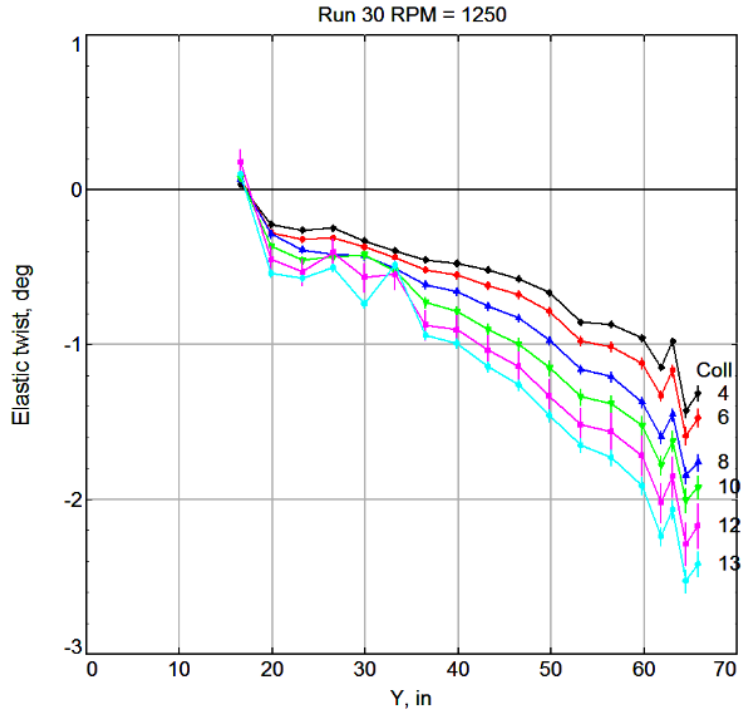
- **Typical photogrammetry measurements of out-of-plane bending (DZ) versus radius (y) and elastic twist vs radius. Each measurement is the average of 128 once-per-rev cycles. Measurements are shown for a collective sweep between 4 and 13 degrees at 1250 RPM.**



Blades with retro-reflective targets



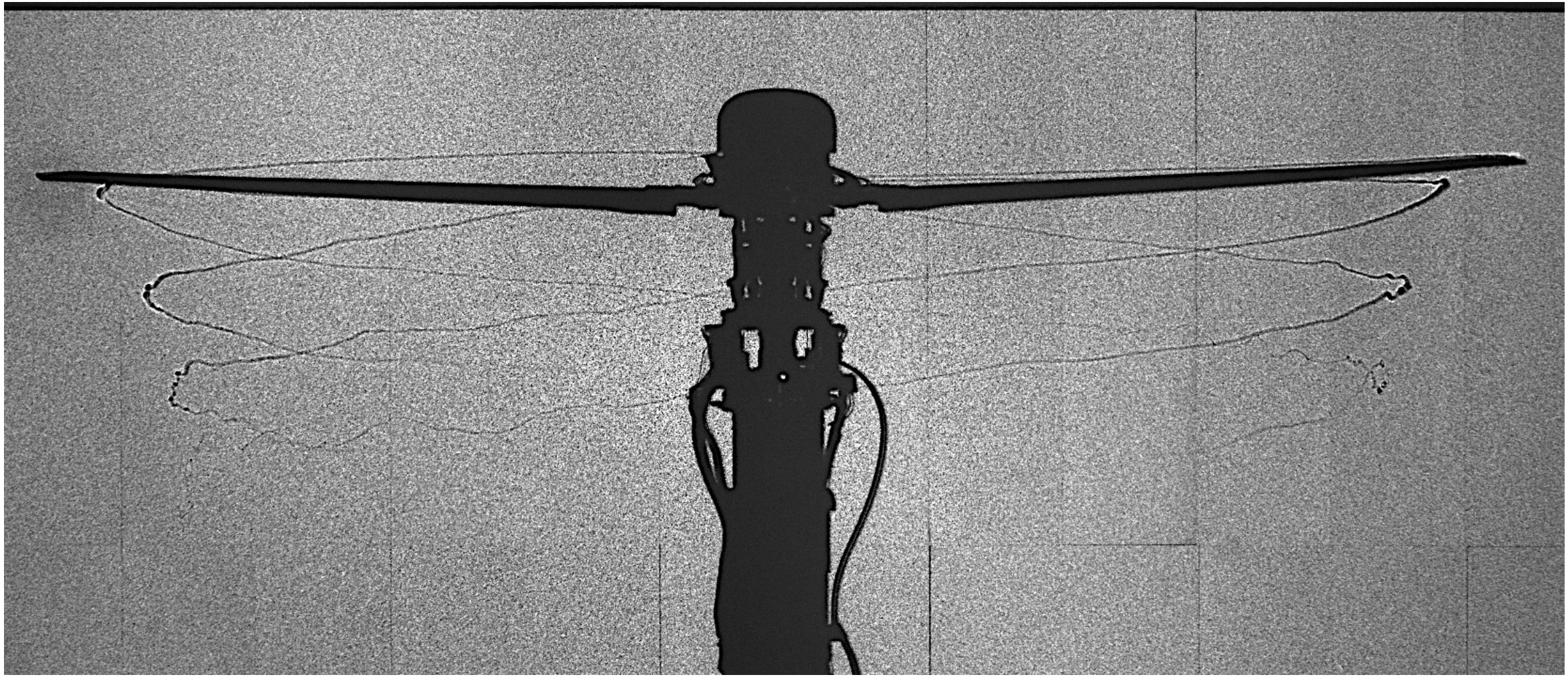
Blade bending (in) vs radius



Elastic twist (deg) vs radius

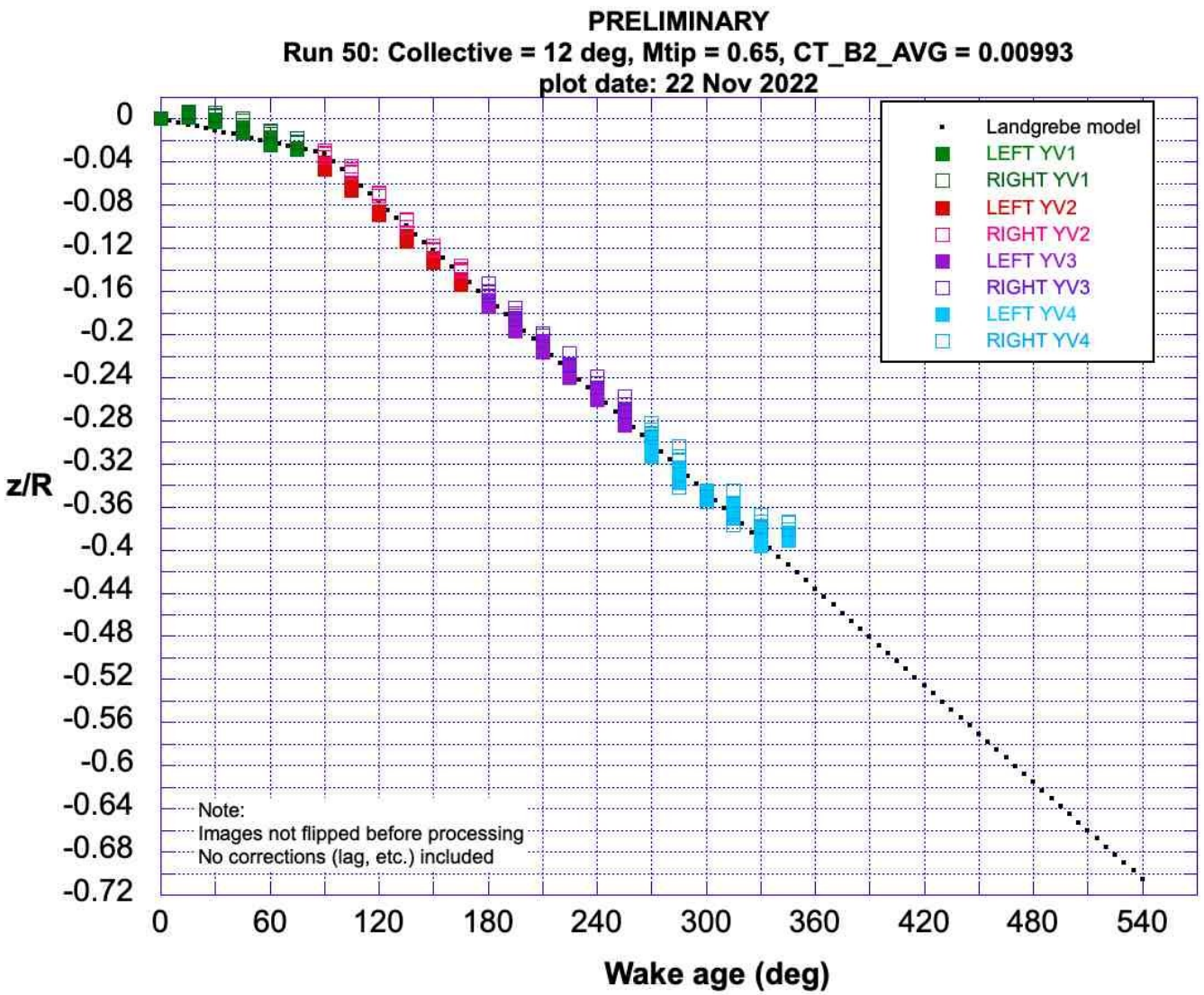
Sample Data – Wake Geometry (Shadowgraph)

Sample shadowgraph image of HVAB rotor (Mtip = 0.65, 12 deg collective) showing rotor tip vortices



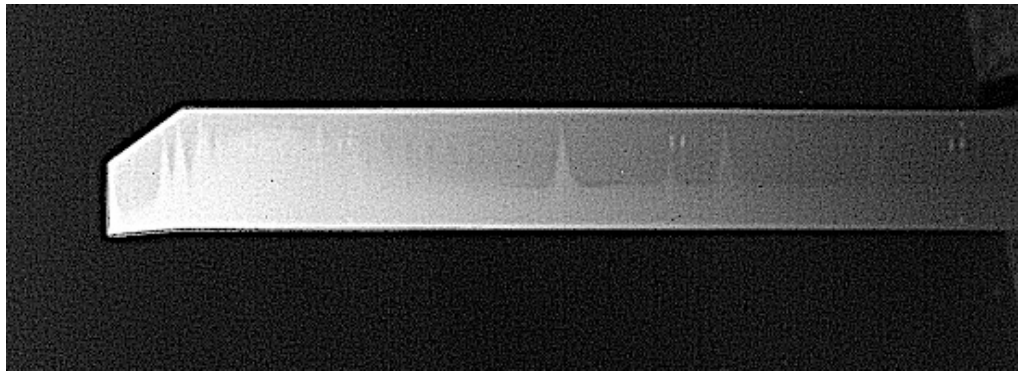
Sample Data – Wake Geometry (Shadowgraph)

Tip vortex vertical descent (Mtip = 0.65, 12 deg collective) vs wake age (Landgrebe model included)

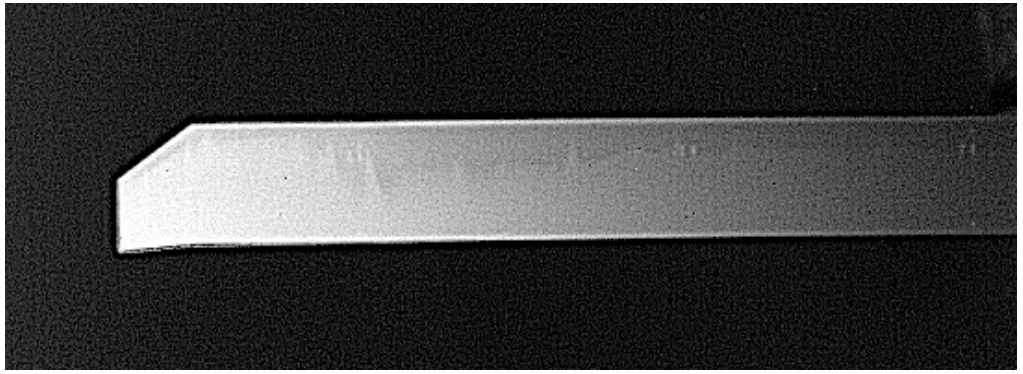


Sample Data – Upper Surface Transition (IR Thermography)

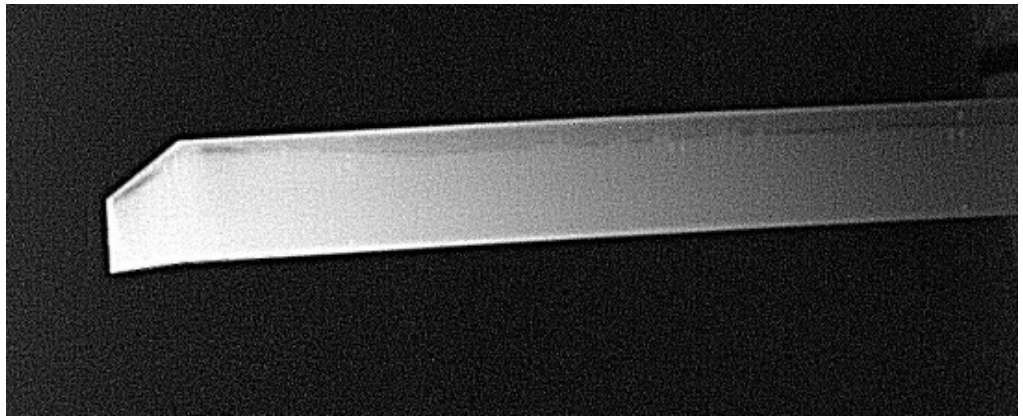
Collective sweep at Mtip = 0.65. Darker colors show laminar flow, lighter colors are turbulent.



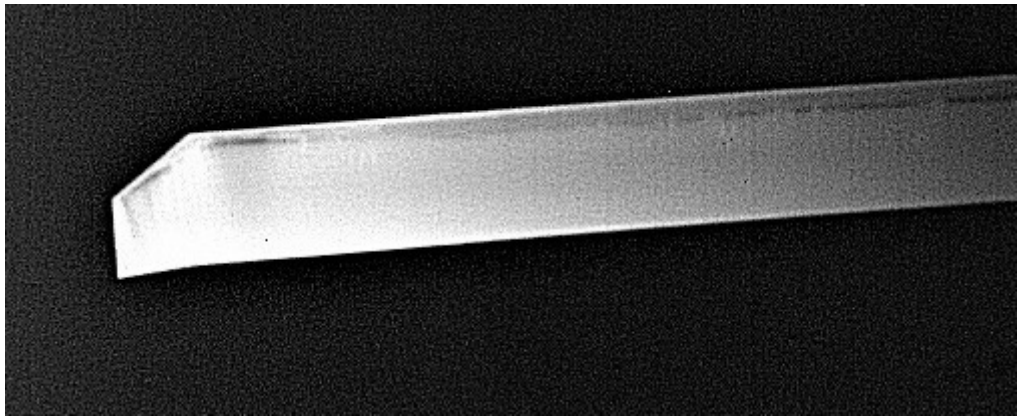
1.8 deg col



6 deg col



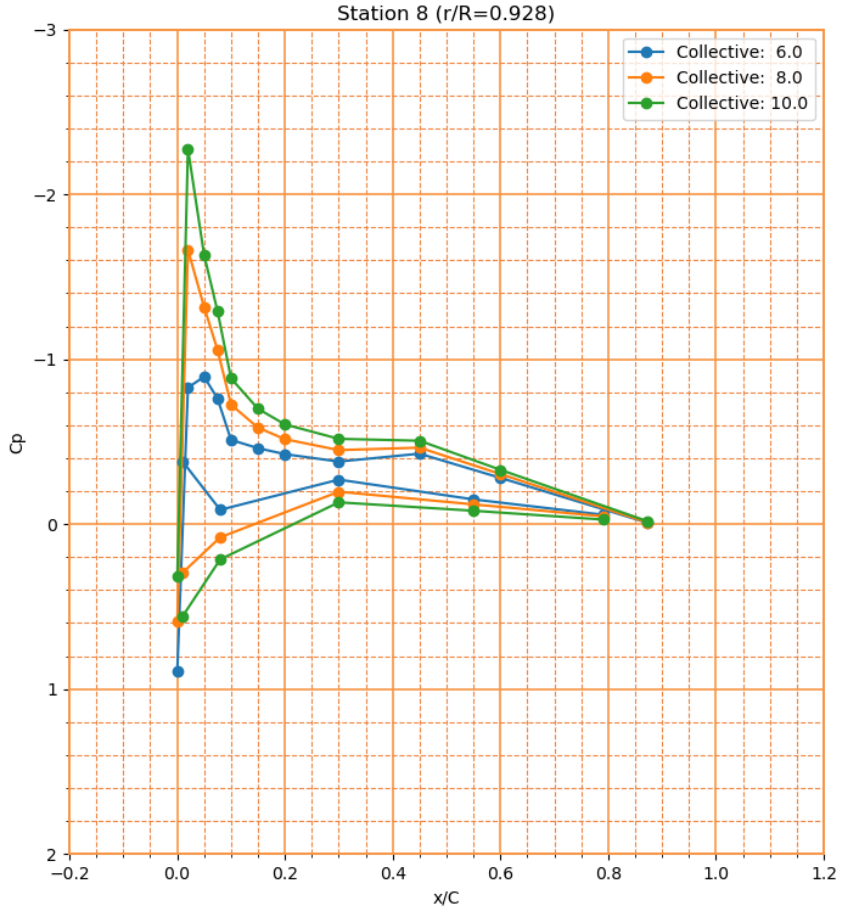
10 deg col



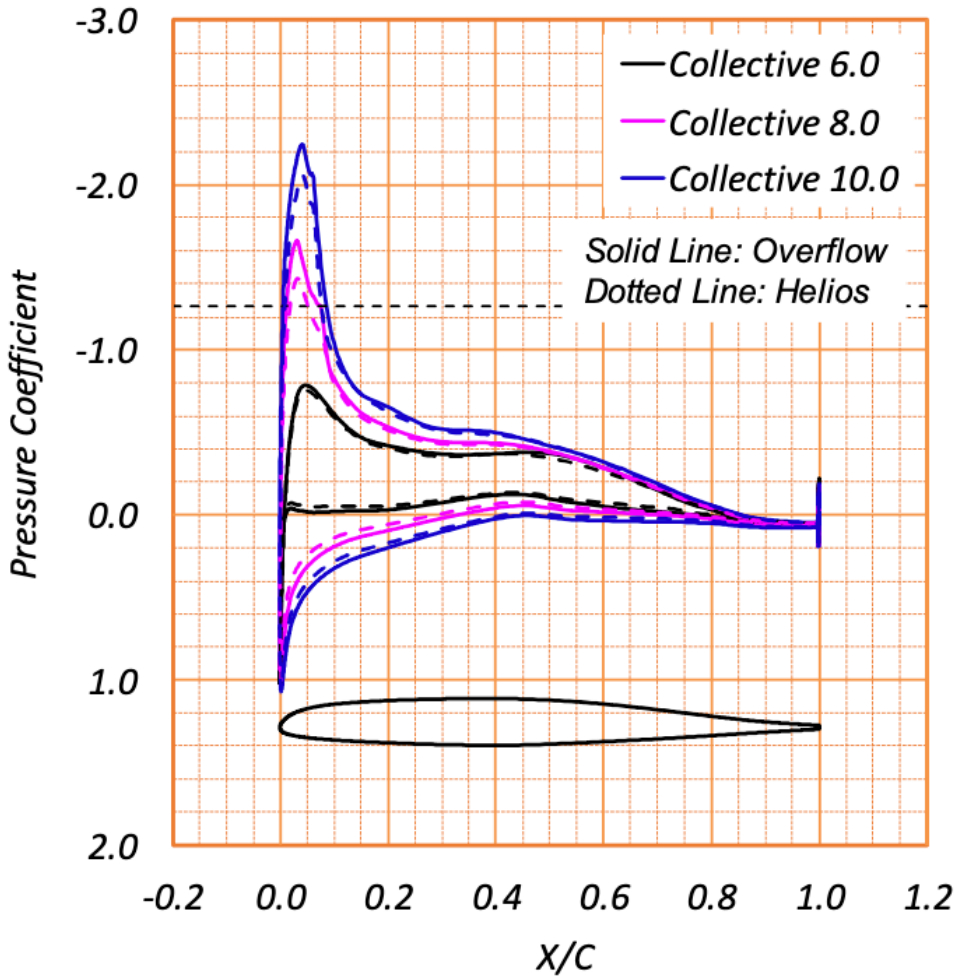
12 deg col

Sample Data – Blade Pressures (r/R = 0.928)

- **Sample Cp data vs x/c for 3 collectives at Mtip = 0.60. Compared with predictions at Mtip = 0.65.**



Test Data at Mtip = 0.60



Predictions at Mtip = 0.650
(Narducci et al, AIAA 2021-0617)

Summary

- **Successfully conducted hover testing of HVAB rotor in NFAC 80x120**
  - Data review and analysis is underway (still lots to do)
  - Some additional performance validation and/or wake characterization (PIV) testing is under consideration

- **Test results to be presented at VFS Forum in May**

- **Goal is to have validated data ready to distribute in May/June time frame**

